



Generative artificial intelligence augmenting SME financial management

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ABSTRACT

This study investigates the potential for entrepreneurs to leverage advances in technological innovation, specifically generative Artificial Intelligence (AI), to build management capability to mitigate business and financial risks. Drawing on theories of Technology Affordances and Constraints and the Resource-Based View (RBV) of the firm, recognising that small and medium-sized enterprises (SMEs) are inherently resource-constrained. We examine how AI-generated financial diagnostics can empower SMEs by generating accessible, real-time analysis and insights, thus bolstering the management function and increasing chances of survival and growth. Using a dataset of 1,150 UK SMEs spanning eight years of financial statements, we test a large language model (LLM) prediction assessment and analyse the potential for SMEs to utilise the technology, notwithstanding enterprise-specific constraints. We conclude that AI may be a very effective tool for smaller enterprises to augment the financial management function, although its efficacy hinges on organisational readiness, competence in interpreting data, and the will to act on automated red-flag alerts. These findings offer practical guidance for SMEs seeking to enhance their financial management processes in today's digital era.

1. Introduction

SMEs are widely recognised as the backbone of global economic growth, serving as critical engines of innovation, employment, and regional development (World Bank, 2022). Likewise, SMEs are significant to the UK, accounting 99 % of all business and a substantial share of employment and private sector turnover (Department for Business Energy & Industrial Strategy, 2022). Despite their pivotal role in fostering economic dynamism and resilience, SMEs face a host of persistent challenges, the most significant of which is a shortage of resources (Levontin et al., 2015). These firms often operate under severe constraints, lacking sufficient financial capital, specialised human resources, and advanced technological infrastructures that are essential for robust financial management and sustained competitive performance (Eisenhardt and Schoonhoven, 1996; Wiklund and Shepherd, 2003).

Resource constraints in SMEs are multifaceted. Not only do these firms struggle with limited access to funding and investment, but they also face deficits in specialised skills, particularly in critical areas such as accounting and financial management (Karadag, 2015). Unlike larger organisations that can afford dedicated finance departments and

external expert consultations, many SMEs rely on ad hoc decision-making and rudimentary record-keeping (Sibanda and Manda, 2016). This resource paucity leaves them particularly vulnerable to financial mismanagement, delayed recognition of liquidity issues, and the risk of going concern failures, a situation where a firm is no longer able to meet its financial obligations (Beck et al., 2008; Lussier and Halabi, 2010).

The Resource-Based View (RBV) of the firm, which states that sustainable competitive advantage arises from possessing valuable, rare, inimitable, and non-substitutable resources (Barney, 1991), further highlights the competitive disadvantages faced by SMEs. Without access to strategic assets, these firms struggle to respond swiftly to market shifts, innovate effectively, or scale operations to meet emerging challenges. This inherent disadvantage is exacerbated in an increasingly complex and volatile economic environment, where timely and accurate financial decision-making is critical for survival. When viewed along with the technology affordance view, we test the concept that access to a new technology can enable entrepreneurs to overcome these resource constraints and gain competitive advantages.

In recent years, however, rapid technological advancements have begun to offer promising solutions to these entrenched challenges

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(Hanelt et al., 2021; Ramdani et al., 2022) Digital affordance theory provides a useful lens through which to examine the potential of such innovations. Digital affordance refers to the practical benefits and opportunities that digital technologies offer (Autio et al., 2018), such as enhanced speed, efficiency, and ease of use, that enable individuals and organisations to perform tasks more effectively than with traditional methods (Majchrzak and Markus, 2013). Essentially, these affordances represent the “action possibilities” that technology makes available, which can help bridge the gap between existing resource limitations and the demands of modern business practices.

One of the most exciting technological developments in this context is the emergence of LLMs, such as ChatGPT. LLMs are advanced artificial intelligence systems that have been trained on vast amounts of textual data, enabling them to understand, predict, and generate human-like language. Their capabilities extend far beyond simple text generation; they can analyse complex patterns in language and data, summarise key insights, and even detect subtle signals within large datasets. We utilise ChatGPT, an advanced LLM developed by OpenAI, due to its accessibility, scalability, and high-quality response generation. Unlike domain-specific AI models, ChatGPT can synthesise diverse financial data, generate predictive insights, and provide natural language explanations, making it particularly suited for SME financial management (Dowling and Lucey, 2023). Its rapid adoption, reaching a million users within a week, further highlights its disruptive potential in AI-driven decision-making (Grant and Metz, 2022). Additionally, beyond the high-level performance in interpretability and versatility demonstrated by Albadarin et al. (2024) and Wen and Chu (2025), ChatGPT’s extensive developer ecosystem and broad user adoption make it especially accessible for smaller enterprises seeking to integrate AI-based solutions. Meanwhile, the entity matching success reported by Nuntachit and Sugannasil (2024), where ChatGPT surpassed BERT and RoBERTa, further reinforces its reliability across various natural language processing tasks. These combined factors underscore why ChatGPT was selected as the most suitable and practical GenAI model for this study. These attributes position ChatGPT as a practical tool for financial risk assessment in SMEs, bridging the gap between complex AI systems and real-world business applications.

For SMEs that lack the resources to maintain full-time financial management teams, LLMs represent a transformative digital affordance, one that promises to democratise access to sophisticated financial analytics. The potential applications of LLMs in SME financial management are significant. These AI systems can rapidly process and interpret large volumes of financial data, identifying patterns and red flags that might otherwise be overlooked by resource-strapped management teams. For instance, an LLM-based tool could be utilised to perform real-time analysis of financial statements, flagging potential liquidity shortfalls, debt misalignments, or other indicators of financial distress. Such early-warning capabilities are particularly valuable for SMEs, where the absence of dedicated accounting functions means that critical financial signals might not be recognised in time to avert crises. Studies in digital entrepreneurship points to the potential of new technologies to help firms pivot rapidly and adapt (Nzembayie and Coghlan, 2024; Urbano et al., 2024) and can serve as an avenue for adaptation and survival (Zahra et al., 2023).

Despite emerging evidence of AI’s potential in corporate finance (Zhang et al., 2023) and going-concern assessments (Mai et al., 2019), there remains a gap in understanding the real-world efficacy of generative AI in SMEs. Firms that typically are unable to leverage advanced data analytics due to resource limitations (Pan et al., 2017). Existing AI-enabled SME risk studies have largely focused on traditional machine learning approaches that require substantial technical expertise and robust data infrastructures, often absent in SMEs. By contrast, generative AI models like ChatGPT offer user-friendly interactions and broader applicability, yet their empirical effectiveness in SME financial diagnostics remains underexplored. To address this gap, we pose the following research question at the core of our study: What is the efficacy

of AI-driven financial management tools, such as ChatGPT, in improving risk assessment and decision-making for SMEs? This is the fundamental contribution of our paper in advancing both theoretical frameworks and practical understanding of AI’s role in SMEs.

Given the strategic importance of early risk detection and effective financial management for SME sustainability (Lukason and Laitinen, 2019), this study seeks to test the central proposition of technology affordance view, by developing an LLM-based tool that provides analytics-based information and analysis which can be used by entrepreneurs and firms to assist them to better understand and manage firm risk, specifically financial risk, thus greatly improving the management function. Specifically, we develop and evaluate a financial management model powered by ChatGPT, aiming to determine whether its analytical outputs can reliably identify going concern issues in SMEs. By integrating insights from both digital affordance theory and the RBV, our research seeks to establish a robust theoretical framework that explains how digital innovations can compensate for resource deficits and enable resource-poor firms to achieve a competitive edge.

This study contributes to the literature in several ways. First, it addresses a notable gap highlighted by Nambisan (2017) regarding the underexplored role of digital technologies in entrepreneurial pursuits. By empirically testing a LLM model on real-world data, our research extends the burgeoning literature on AI and provides new evidence on the practical utility of LLMs in financial risk management for SMEs. Second, we advance theoretical development by integrating the RBV (and related resource deficit theory) with the technology affordances and constraints perspective (Majchrzak and Markus, 2013; Gibson, 1977). This combined framework offers a more robust explanation of how affordable digital technologies can empower resource-constrained firms to overcome inherent limitations, thereby creating competitive advantages in dynamic market environments. Ultimately, our study demonstrates that the democratisation of sophisticated information systems through accessible AI solutions not only transforms entrepreneurial decision-making but also levels the playing field for smaller enterprises competing against better-resourced rivals.

The remainder of this paper proceeds as follows: in the next section we review the relevant literature, followed by a detailed description of our methodology, presentation of empirical results, and discussion of broader theoretical and practical implications. Finally, we conclude with proposed directions for future research into AI-driven innovation for SMEs.

2. Theoretical context and previous related literature

This study is framed within a dual-theoretical perspective, integrating RBV and technology affordance and constraint theory. We apply these frameworks to SME financial management and the adoption of LLMs. RBV helps to explain the resource limitations faced by SMEs, while technology affordances and constraints theory highlights the opportunities and challenges presented by digital innovations.

2.1. Digital entrepreneurship and AI solutions

The emergence of digital entrepreneurship offers a chance to transform traditional models of SME finance and risk management. Nambisan (2017) describes how digital infrastructures reduce organisational boundaries, enabling continuous innovation and more dynamic entrepreneurial processes. AI technologies enable machines to learn from experience, adapt to new inputs, and perform human-like tasks, enhancing efficiency and decision-making across various industries (Liu et al., 2020). Studies observe that advanced data analytics and AI can fundamentally change how smaller firms conduct financial analyses, making sophisticated predictive methods feasible even with limited internal expertise (Mai et al., 2019; du Jardin, 2021). Specifically, LLMs such as ChatGPT allow for user-friendly interaction with large datasets, potentially alleviating SMEs’ reliance on specialised technical teams

(Mollick and Euchner, 2023). However, existing research reveals how constraints, ranging from data quality to ethical and interpretability concerns, can undermine the benefits of AI for SMEs (Berente et al., 2021). As the complexity of AI-driven systems grows, entrepreneurs often lack the necessary digital literacy to navigate black-box outputs or ensure consistent data inputs (Maragno et al., 2023). These shortfalls mirror prior findings that technology alone cannot guarantee improved managerial decisions unless organisational practices adapt accordingly (Freel, 1999; Carter and Van Auken, 2006).

Recent advances in digital entrepreneurship highlight how AI can reshape entrepreneurship by offering data-driven insights and enhanced decision-making (Giuggioli and Pellegrini, 2023; Obschonka and Audretsch, 2020). Studies increasingly view AI as a super-tool augmenting entrepreneurs' creativity, opportunity recognition, and strategic decision-making (Shepherd and Majchrzak, 2022). For instance, AI can handle vast, unstructured data to help discover and exploit entrepreneurial opportunities under uncertainty (Chalmers et al., 2023), while also transforming how ventures are created, scaled, and governed (Chalmers et al., 2021). These developments align with studies showing that SMEs and other early-stage ventures often rely on subjective judgments and limited information when assessing risk highlighting the need for more robust analytic tools (Elisa et al., 2021; Manigart and Khosravi, 2024). AI-driven methods, such as machine LLMs, enable the processing of vast datasets while reducing human biases in model selection (Fosso Wamba and Akter, 2015; Bukovina, 2016). For instance, emerging research shows that LLMs can interpret market news or identify hidden signals (Xie et al., 2023), illustrating how they might similarly detect signs of financial vulnerability in smaller firms.

Although several studies have demonstrated the promise of AI in capital markets and corporate forecasting (Zhang et al., 2023), few address the unique context of SME finance or going-concern assessments. Prior work on SME risk prediction has largely employed machine learning to evaluate venture outcomes (Ross et al., 2021), but these methods generally require specialised technical skills and comprehensive datasets, factors that can hamper widespread adoption by resource-constrained SMEs (Alaka et al., 2018; Pan et al., 2017). Despite these challenges, generative AI tools like ChatGPT offer a user-friendly alternative, potentially lowering adoption barriers. Studies by Liu et al. (2023) and Mollick and Euchner (2023) emphasise how generative AI can streamline business tasks and improve entrepreneurial decision-making, yet there is limited empirical research on applying ChatGPT specifically to SME financial challenges. Consequently, examining the technology's affordances and constraints in early-stage or smaller-scale contexts remains a critical research frontier for digital entrepreneurship (Autio et al., 2018; Petzsche et al., 2023; Nambisan, 2017; von Briel et al., 2018).

2.2. Technology affordances and constraints theory in the SME context

Technology affordances and constraints theory (Gibson, 1977; Majchrzak and Markus, 2013) offers a helpful framework for our study. According to Majchrzak and Markus (2013, p.832), "technology affordances represent the 'action potential', that is, to what an individual or organisation with a particular purpose can do with a technology or information system; technology constraint refers to ways in which an individual or organisation can be held back from accomplishing a particular goal when using a technology or system".

Effectively, within technology affordance and constraints theory, affordances represent the potential actions a technology's features make available, whereas constraints indicate barriers that limit the user's objectives (Leonardi, 2011; Autio et al., 2018). In SME financial management, LLMs might afford real-time risk assessments or liquidity forecasts, yet also constrain users if data standardisation or interpretability remains problematic (Maragno et al., 2023). Nambisan et al. (2019) highlight that digital technologies' impacts vary greatly depending on how entrepreneurs perceive and utilise these affordances.

These technological affordances will only be realised by aligning business models to technological requirements in order to overcome their respective constraints (Baden-Fuller and Haefliger, 2013). The same platform that stimulates collaborative problem-solving in one setting may generate confusion in another if user goals and capabilities fail to align with the platform's features (Evans et al., 2017). According to Majchrzak and Markus (2013) understanding a technology's use and its outcomes depends on examining how individuals and organisations interact with the technology. A "technology affordance" emerges when a particular technology enables certain actions that people or organisations can perform, whereas a "technology constraint" describes how the same technology may hinder them from achieving a goal. A simple example is provided by Von Briel et al. (2018) and Du et al. (2019) who both demonstrate how digital tools improve efficiency while simultaneously adding a level of complexity and a need for planning to unlock their affordances. Similarly, Leonardi's (2011) study demonstrates how perceived constraints can drive adaptation: SMEs might try to recalibrate how they input data or interpret model outputs, therefore reshaping both the technology and their routines. Understanding these interactions is crucial, as SMEs with resource shortages cannot always undertake the systemic changes needed to fully realise an AI tool's affordances (Berente et al., 2021).

Technology affordance and constraint theory frames technology adoption as a process of dynamic engagement among users, tools, and organisational contexts, particularly suitable for explaining LLM usage in SME financial risk management, where actors face both enabling capabilities (e.g., automated red-flag detection) and pressing constraints (e.g., budget, skill, or data concerns). In addition to this, the use of big data offers a level of efficiency but will need effective tools in place to streamline and maximise value (Lehrer et al., 2018). A similar finding was made by Trocin et al. (2021) when analysing how HR firms found greater fairness in hiring when using AI, however these affordances were constrained by ethical and regulatory requirements. Several studies have explored the affordances and constraints of emerging technologies. For instance, Parolin et al. (2023) illustrate how sustainability-focused tools can enhance innovation but also encounter limitations that reduce their overall effectiveness. Likewise, Lee (2023) demonstrates that the strategic integration of converging technologies can drive innovation, albeit with challenges in effectively merging these tools. In addition, An and Rau (2019) examine how AI and blockchain adoption can unlock new business opportunities while revealing inherent constraints that necessitate organisational change. Steininger et al. (2022) further emphasise the need for firms to adapt their traditional approaches to fully leverage the potential of these evolving technologies.

By focusing on these relational concepts, researchers can better identify the conditions under which digital entrepreneurship tools become transformative or remain underutilised. Advancing knowledge on how SMEs can harness AI to mitigate going-concern issues. In turbulent times and crises, recent research highlights the necessity of technological innovations for addressing emerging challenges (Lee, 2023), while also highlighting dynamic capabilities, particularly innovation, digitalisation, and collaboration, are crucial for ensuring SME resilience (Vasi et al., 2024). These strategic capabilities are integral to effective SME financial management, enabling agile resource allocation, risk mitigation, and sustained financial stability amid ongoing uncertainty, of which are becoming increasingly common (Kaya, 2022; Valdez-Juárez et al., 2024).

The theoretical understanding of digital entrepreneurship remains limited, offering insufficient insights into how it enhances resilience both at organisational and societal levels (Nambisan, 2017; Kraus et al., 2019; Beliaeva et al., 2020). Our study seeks to advance technology affordance and constraint theory by examining how digital artefacts, particularly LLMs, can strengthen SME financial management and drive organisational resilience. In doing so, we propose the development of an AI-enabled management tool specifically designed to address the financial complexities of SMEs. This approach not only deepens the

theoretical discussion on digital entrepreneurship but also provides a practical framework for leveraging emerging technologies to bolster SMEs' adaptive capacities.

2.3. SME resource based view

This RBV theory has been widely employed to explain why certain firms outperform others in competitive markets and how internal assets, ranging from tangible capital to intangible knowledge, drive performance (Grant, 1996). RBV states that competitive advantage is rooted in the unique resources and capabilities that a firm possesses (Barney, 1991; Peteraf, 1993).

When applied to SMEs, the RBV framework reveals a distinct challenge. SMEs often operate under chronic resource constraints, be it in financial capital, skilled personnel, or technological infrastructure, that undermine their capacity to build and sustain competitive advantages (Hitt et al., 2001). This resource deficiency not only limits their ability to implement sophisticated strategic initiatives but also predisposes them to risk-averse behaviours that can stifle innovation and growth.

Studies have further elaborated on the implications of resource scarcity for SMEs. Levontin et al. (2015) illustrate that acts of "emptying" trigger perceptions of resource deficiency, leading to heightened self-focus and short-term survival tactics at the expense of long-term investments. Similarly, Van Burg et al. (2012) demonstrate that resource constraints can skew opportunity identification in SMEs. They argue that when faced with pressing operational issues, such as supply or demand challenges, SMEs may overlook larger growth and innovation prospects, thus reinforcing a cycle of short-sighted decision-making. Robson et al. (2013) extend this line of inquiry by highlighting how resource deficits manifest in credit-rationing phenomena. Their findings suggest that while experienced entrepreneurs may secure better financing, innovative but resource-poor SMEs often struggle to access the capital necessary for expansion due to lenders' risk aversion. Complementing these insights, Grable et al. (2013) provide evidence that subjective perceptions of income inadequacy can further exacerbate financial decision-making challenges, leading SME managers to rely on intuition over data-driven approaches. These studies highlight that resource constraints in SMEs are not solely a matter of objective scarcity but also of perceived insufficiency.

2.4. SME financial management

SMEs play an essential role in driving innovation and employment (McMahon, 2001; Chege and Wang, 2020) yet they experience disproportionately high failure rates (Carter and Van Auken, 2006; Lukason and Laitinen, 2019). Common exposures include weak internal controls, undercapitalisation, and poor cash-flow oversight (Marriott and Marriott, 2000). Such practices often prevent SMEs from detecting early signs of distress, further hindering their ability to secure timely financing (Ciampi and Gordini, 2013). While established finance theories highlight capital structure decisions (Robb and Robinson, 2014; Degryse et al., 2012), SMEs' constrained resources and lack of sophisticated financial infrastructures exacerbate their exposure to liquidity shocks (Carpenter and Petersen, 2002; Kaya, 2022).

Financial stress can escalate into going-concern uncertainties if unresolved, particularly in turbulent environments (Cole and Damm, 2020). Research on going-concern prediction, especially for smaller firms, highlights the need for accurate early-warning systems (Martens et al., 2008; Chen, 2019). Studies have long shown that conventional prediction models often focus on large corporations (Mai et al., 2019), leaving a gap in practical solutions that cater to resource-constrained SMEs (Fasano et al., 2024).

SMEs have relied on their accounting practitioners not only for core reporting requirements (Nigri and Del Baldo, 2018) but also for consultancy and advice, particularly when implementing new initiatives and processes (Jarvis and Rigby, 2012). Several studies emphasise the

importance of these practitioners, frequently described as SMEs' "most trusted advisor" (Blackburn et al., 2010; European Federation of Accountants and Auditors for SMEs, 2018; Spence et al., 2012). As SMEs navigate a range of uncertainties, business owners are increasingly turning to their trusted advisors for guidance to secure long-term viability. Yet 80 % of SMEs do not employ a full-time accountant (Berthelot and Morrill, 2016) a gap that poses critical challenges in spotting real-time financial issues or looming going concern threats. This shortage of in-house expertise means that many owners cannot readily access detailed, up-to-date financial counsel, often leading to delayed or ad hoc responses to emerging problems (Berthelot and Morrill, 2016). Consequently, SME resource constraints reinforce their dependence on external accountants, heightening the risk that financial vulnerabilities go undetected until they escalate.

Considering these challenges, an AI-driven tool capable of promptly detecting red flags can provide a vital complement to the limited financial expertise within many SMEs, especially in circumstances where on-demand professional guidance is not readily available. By offering real-time analysis and predictive insights, AI, if properly trained and aligned with practical user needs, could help close the resource gap that increases SMEs' exposure to financial distress and going-concern risks.

2.5. Dual-theoretical perspective

RBV highlights that firms derive sustainable competitive advantage from possessing resources, such as specialised knowledge or proprietary technologies, that are valuable, rare, inimitable, and non-substitutable (Barney, 1991; Peteraf, 1993). However, SMEs are often constrained by limited financial capital, skilled personnel, and technological infrastructure (Hitt et al., 2001), making it difficult to invest in sophisticated analytical tools. This resource gap not only fosters risk-averse strategies (Levontin et al., 2015) but can also deter innovation, as firms may overlook growth opportunities in favour of short-term survival (Van Burg et al., 2012; Robson et al., 2013; Grable et al., 2013).

Subsequently, technology affordances and constraints theory (Gibson, 1977; Majchrzak and Markus, 2013) suggests that digital tools can provide "action possibilities" enabling firms to overcome certain operational limitations, provided they can effectively harness the technology's features. For instance, a well-designed LLM can afford rapid detection of going-concern issues (e.g., liquidity or solvency red flags) if the SME can supply structured financial data and interpret the outputs. Yet the very constraints that plague SMEs, limited budgets, technical skill deficits, or inconsistent data, may also hinder them from realising these affordances (Berente et al., 2021). Therefore, the key question becomes under what conditions a resource-scarce SME can successfully activate the AI affordances that RBV flags as critical for achieving competitive advantage.

Drawing these perspectives together reveals a dynamic connection between resource scarcity (RBV) and affordance activation (technology affordances). From the RBV viewpoint, even a small infusion of external expertise (e.g., via accounting advisors or minimal IT support) can transform intangible assets (such as SME managerial insight) into a valuable resource base for operationalising AI tools. Meanwhile, technology affordances can reduce the firm's dependence on in-house analytics skills by offering user-friendly interfaces or automated diagnostic features. However, as Nambisan et al. (2019) emphasise, the benefits of digital tools hinge on entrepreneurs' ability to perceive, adopt, and exploit these technologies effectively. In other words, resource-constrained SMEs succeed in realising AI affordances when they can identify a concrete managerial need (e.g., timely liquidity checks) that aligns with the AI tool's capabilities. This is especially vital in context like going-concern assessments (Lukason and Laitinen, 2019). If firms secure minimal but sufficient complementary resources, be it external guidance or basic IT infrastructure, to facilitate adoption and data quality; without such foundational support, many SME digital

initiatives falter (Pan et al., 2017). To develop or access digital literacy among key decision-makers, enabling them to interpret automated signals and integrate them into broader strategic processes (Freel, 1999); and to address organisational and behavioural constraints, ensuring that employees remain open to algorithmic input rather than reverting to ad hoc or intuition-based decision-making (Carter and Van Auken, 2006).

In sum, RBV clarifies why SMEs struggle to implement advanced data analytics, due to resource deficits, while technology affordances and constraints theory highlights how, given certain supportive conditions, digital tools like LLMs can partially alleviate these limitations. The synergy between these theoretical lenses highlights the nuanced reality that AI neither automatically solves resource deficiencies or remains out of reach for small businesses; instead, SMEs must actively cultivate the minimal organisational, financial, and technological capacity required for AI-driven insights to take hold. The dual-theoretical framework provides a comprehensive lens for understanding how SMEs can leverage emerging digital tools to overcome inherent resource limitations and enhance their financial management capabilities, thereby reducing vulnerability to going-concern risks. This study thus examines how generative AI tools, such as ChatGPT, might help SMEs convert limited resources into an adaptive capability for real-time financial risk management, highlighting both the promises and the persistent constraints that shape technology adoption in entrepreneurial contexts. Fig. 1 below shows the theoretical interlink of our study.

This study empirically evaluates the effectiveness of LLM-based financial analysis in enhancing decision-making processes, bridging the gap between theoretical affordances and practical SME adoption.

3. Data and methodology

3.1. Data

Our study focuses on 1,150 SMEs in the United Kingdom, all of which were incorporated between 2014 and 2016. We utilise an eight-year observation period for each firm, drawing on the FAME database, a Bureau van Dijk platform that provides detailed financial accounts, balance sheets, and key ratios. FAME is a well-known and reputable source for entrepreneurial finance research that been used in multiple studies within entrepreneurial finance and predictive modelling for SMEs (Cheng et al., 2024; O'Reilly et al., 2025). After cleaning the dataset, our final sample contains firms that are private, for-profit SMEs meeting the standard criteria for UK SME classification and have clean consecutive years of financial statement data. To construct a

comprehensive financial overview of each firm, essential variables such as revenue, net profit, total assets, liabilities, liquidity, and gearing were gathered. We also ensure that a certain number of firms received a going-concern warning from auditors after the initial four-year observation period and firms that subsequently failed are also included. We assess the initial four years of financial statement data and compare with the actual outcomes in years five to eight. In total, 320 of the 1,150 SMEs received going-concern warnings, reflecting a significant subset at heightened risk of financial distress. In addition to financial data, we also include non-financial data such as sectoral classification and location. The purpose of developing such a comprehensive dataset is to assess the accuracy of ChatGPT at developing a scoring system for firms that are potential going concern warnings. Variable definitions are presented in Table 1. It should be noted, we developed additional financial variables that have been used extensively in SME finance literature such as total debt to total assets (Demirgüç-Kunt et al., 2015; 2020), interest coverage (Vander Bauwhede et al., 2015) and liquidity thresholds (O'Reilly et al., 2021). Information on the company profiles, including sector and location, are included in Table 2. The summary statistics for financial data used for prompting ChatGPT and used in our statistical model are summarised in Table 3.

3.2. Methodology

We aim to evaluate whether the advanced reasoning capabilities of the ChatGPT o1 model can effectively assist in identify potential red flag warnings for SMEs. ChatGPT o1 and ChatGPT-4 are both tools from OpenAI, however o1 reasoning is designed for solving complex problems which requires deeper prompts, reasoning and analysis. This makes it more suitable to SME finance functions compared to ChatGPT-4 which is better suited to creativity and broader applications.

By employing ChatGPT o1 reasoning as our AI-based screening tool, we undertake a probability-like assessment of each firm's financial health, transforming four consecutive years of historical financial data into a concise 0–100 stability score. The primary objective is to determine whether ChatGPT can effectively flag at-risk SMEs, namely, those likely to receive going-concern warnings, by generating a “red flag” rating for scores below 30. In doing so, we aim to evaluate how well ChatGPT complements or substitutes for traditional early-warning methods in SME financial management.

First, we anonymise all firms by assigning unique identifiers, ranging from Company 1 to Company 1,150. For each company, we prepared a standardised prompt containing four years of financial statements

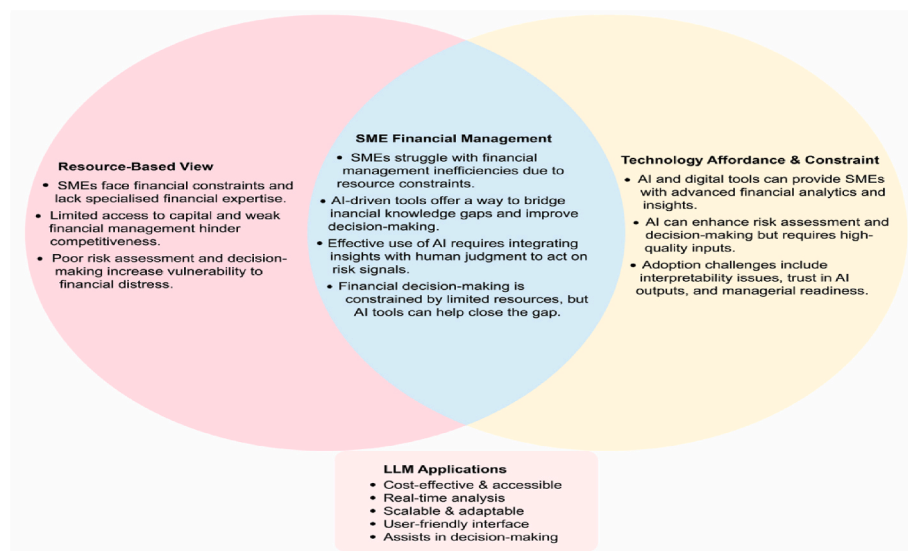


Fig. 1. Theoretical framework.

Table 1
Variable definitions.

Variable	Definition
Going Concern Warning	Dummy = 1 if a firm received a going concern warning from auditors
LLM Score	Score that LLM provided to individual company on a scale of 0–100
LLM Red Flag	Dummy = 1 if a firm was classified as red flag warning (LLM score 0–30)
Total Assets	Log amount of total assets (proxy for firm size)
Revenue	Log amount of sales (£) in given financial year
Revenue (YoY Growth)	Year-on-year revenue growth expressed in %
Profitability	Log amount of earnings before interest, tax, depreciation and amortisation (£) in given financial year
Profit Margin	Profit margin expressed in %
Retained Earnings	Log amount of retained earnings (£) at financial year-end
Firm Age	Number of years since incorporation
Gearing	Gearing ratio (debt/equity) at financial year-end
Gearing_095	Dummy = 1 if a firm has a gearing ratio of greater than 0.95:1
Liquidity	Liquidity ratio (current assets/current liabilities) at financial year-end
Liquidity_075	Dummy = 1 if a firm has a liquidity ratio of less than 0.75:1
Debt/Total Assets	Debt/total assets at financial year-end
Total Liabilities	Log amount of total liabilities
Interest Coverage	Interest cover in times (earnings before interest, tax, depreciation and amortisation/interest)
Bank	Log amount of bank balance (£) at financial year-end
Receivable Days	Average number of days it takes for a company to collect payments (trade receivables/revenue x 365)
Payable Days	Average number of days it takes for a company to pay suppliers (trade payables/cost of sales x 365)
Sector	Sector classification: Business and Professional Services, Construction and Property, Consumer Services, Manufacturing, Retail and Wholesale.
Location	Specific location in the United Kingdom: London, South, East, North, West, and Other UK.

Table 2
Company profiles.

Sector	N	%
Business and Professional Services	277	24.09
Construction and Property	175	15.22
Consumer Services	251	21.83
Manufacturing	248	21.57
Retail and Wholesale	199	17.30
Total	1,150	100.00
Location	N	%
London	224	19.48
South	221	19.22
East	198	17.22
West	214	18.61
North	197	17.13
Other UK	98	8.35
Total	1,150	100.00

drawn from the FAME database. This prompt included revenue trends, profitability, liquidity measures, and gearing ratios, ensuring consistent input across the entire sample. We design the scoring metrics and scoring classification (see [Appendix 1 and 2](#)) which are provided to ChatGPT to screen and firm data points were uploaded, ChatGPT produced a single composite score (0–100) reflecting the firm's perceived likelihood of being financial distressed. Scores under 30 were designated as high-risk (red flag).

We then tracked each company's actual status in the four years following our assessment period (T+1 to T+4), noting whether the firm received a formal going-concern warning from auditors. This approach enables us to compare ChatGPT's risk assessment, captured by its scoring system, with genuine post-hoc outcomes. By analysing the extent to which the AI model accurately flags firms that later exhibited

financial problems, we can gauge ChatGPT's capacity to serve as an early-warning mechanism and thereby deepen our understanding of how generative AI might enhance, or even transform, conventional SME risk assessment practices.

Our methodology proceeded in several stages to assess ChatGPT's ability to identify SMEs at high risk of going-concern warnings. First, we evaluated how well the model could distinguish high-risk and low-risk firms using four years of financial data per SME. Following prior research on SME failure, we focused on liquidity, leverage, and profitability as predictive indicators of distress. After feeding each firm's financial statements into ChatGPT, we recorded a "financial stability" score ranging from 0 to 100. Firms scoring below 30 were flagged as high-risk ("red-flag" zone), while higher scores indicated lower concern. We then compared these predictions with actual outcomes, specifically, whether auditors issued going-concern warnings up to four years post-assessment, providing an initial gauge of ChatGPT's effectiveness as an early-warning mechanism.

Next, we developed a probability-style scoring range to capture the model's full risk spectrum for all 1,150 SMEs. ChatGPT's continuous 0–100 score was grouped into four bands, Red (0–30, high risk), Amber (31–50, moderate caution), Green (51–75, stable), and White (76–100, strong), and compared with the actual occurrence of going-concern warnings. This step allowed us to examine how closely ChatGPT's categorical classifications aligned with real-world financial outcomes.

We then assessed predictive accuracy and model effectiveness by generating confusion matrices that matched ChatGPT's predicted risk categories against whether a firm ultimately received a going-concern warning. This approach enabled us to calculate accuracy, precision, recall, and F1-scores, offering quantitative insights into ChatGPT's performance in distinguishing high-risk from low-risk SMEs. We also computed the Brier Score ([Brier, 1950](#)) to measure the accuracy of the model's estimated probabilities, supplemented by a calibration plot illustrating performance across different risk thresholds. To further explore the model's robustness, we performed a Probit regression using ChatGPT's 0–100 score as a primary independent variable, controlling for standard financial ratios. This allowed us to statistically isolate the effect of ChatGPT's assessment on the likelihood of a going-concern warning and to determine whether the AI-based risk signal added value beyond conventional metrics in a Probit regression ([Table 6](#)). Pearson correlation ([Table 7](#)) diagnostics were also conducted to check for multicollinearity in the Probit specification.

By employing this multi-layered assessment, we provide a thorough evaluation of ChatGPT's scoring system. Specifically, we compare its classifications against real-world outcomes to determine whether the model can reliably issue early warnings for SMEs in precarious financial positions. In our sample of 1,150 firms, 320 received going-concern warnings within eight years of trading, offering a substantial test of ChatGPT's ability to detect early signs of distress. Through this process, we aim to clarify how generative AI can assist, complement, or refine existing methods of financial risk assessment within the SME context and build on technology affordance and constraint theory.

3.3. ChatGPT configuration and prompt engineering

To utilise ChatGPT effectively for assessing SME financial stability, we converted each firm's raw numerical data into standardised textual summaries that the model could interpret. Rather than providing isolated numeric fields, we presented coherent written descriptions of four consecutive years of financial statements. This transformation into short paragraphs allowed ChatGPT to analyse quantitative trends in a format aligned with its language-based processing capabilities. We ensured a uniform structure across all firms, consistently describing annual revenue changes, profitability markers, and selected financial ratios. Each summary omitted any details beyond the four-year window to avoid bias and maintained a neutral tone that refrained from leading language or indicating the firm's eventual going-concern outcome.

Table 3
Summary descriptive statistics of financial data.

Variable	N	Mean	Min	Max	Median	Standard Dev
Going Concern Warning	320	0.49	0.00	1.00	0.00	0.50
LLM Score	1,150	55.00	5.00	99.00	51.86	29.16
LLM Red Flag	370	0.32	0.00	1.00	0.00	0.50
Total Assets	1,150	6,756,920	599057	19,313,296	5,405,811	1,958,545
Revenue	1,150	3,110,598	0	9,546,175	2,574,108	2,755,294
Revenue (YoY Growth)	1,150	13.73	(49.26)	101.57	8.31	21.10
Profitability	1,150	510,445	(2,582,084)	5,864,322	254,443	252,952
Profit Margin	1,150	7.02	(19.84)	59.92	5.88	9.41
Retained Earnings	1,150	1,024,508	(6,447,735)	29,300,278	1,302,974	683,694
Firm Age	1,150	4.00	4.00	4.00	4.00	14.42
Gearing	1,150	2.64	0.00	29.64	0.82	1.46
Gearing_095	1,150	0.48	0.00	1.00	0.00	0.50
Liquidity	1,150	2.21	0.00	14.46	2.11	1.12
Liquidity_075	1,150	0.41	0.00	1.00	0.00	0.50
Debt/Total Assets	1,150	0.28	0.00	0.99	0.33	0.19
Total Liabilities	1,150	(4,558,491)	0	(14,101,973)	(2,591,495)	(1,547,041)
Interest Coverage	1,150	3.65	0.00	25.98	2.43	1.31
Bank	1,150	252,037	0	5,477,547	325,975	822,885
Receivable Days	1,150	51.97	0	109.50	48.73	19.08
Payable Days	1,150	56.45	0	119.32	49.44	20.15

Table 4
Confusion matrix.

Outcome	Actual (N)	AI Predicted Going Concern	AI Predicted No Going Concern
Going Concern	320	297 (TP)	23 (FN)
No Going Concern	830	73 (FP)	757 (TN)
Total	1,150	370	780

Table 5
Summary of F1 scores from each category.

Category	Precision	Recall	F1-Score
Going Concern	80.3 %	92.8 %	86.2 %
No Going Concern	97.2 %	91.2 %	94.1 %
Overall Accuracy	91.7 %		
Macro F1-Score	–	–	90.1 %
Weight F1-Score	–	–	91.9 %

All prompts were delivered to ChatGPT programmatically via the OpenAI API, using a standardised script that automatically generated and submitted textual summaries for each SME. This automated process minimised variation in prompt length or emphasis, preserving consistency across the dataset. To mitigate the risk of “hallucinations”, where the model might produce extraneous or fabricated details, we kept prompts strictly factual, describing only four years of audited financials without speculative language or leading questions. We also set ChatGPT’s temperature parameter to 0, reducing randomness and increasing the likelihood of stable, reproducible responses. After the model returned an initial score for each firm, we systematically re-ran a random subset (approximately 10 % of firms – 150 in total) of these prompts to verify that the scores remained consistent under identical input conditions. Minor deviations (± 1 to 2 points) were rare and typically related to slight reformulations ChatGPT might introduce, but no major anomalies or contradictory outputs were observed.

While ChatGPT is a proprietary tool, which poses concerns for reproducibility, future updates or parameter changes may alter its behaviour. To address this, we specify the API version and configuration settings used, and we retain all input prompts, anonymised data, and output logs. Researchers wishing to replicate or extend our work may not achieve the same numerical scores if the underlying model evolves, but our transparent methodology including standard scripting approach, the same 0–100 scoring scale, consistent prompt templates, should

permit a comparable procedure and facilitate external validation. A sample prompt is provided in [Appendix 3](#) for reference.

By implementing these measures that include stringent prompt design, low randomness, repeated scoring checks, and meticulous documentation, we sought to balance the convenience and responsiveness of generative AI with core scientific values such as reproducibility and rigor. While ChatGPT’s “black box” nature cannot be fully eliminated, our standardised process and methodological disclosures aim to maximise transparency and minimise the risks associated with proprietary LLM-based research.

4. Results

This section summarises the results of our study, followed by a discussion on the practical implications of our findings.

4.1. Results

After providing ChatGPT with four consecutive years of financial data for each of the 1,150 SMEs, we compared its risk classification against actual going-concern outcomes observed in Years 5–8. In total, 320 firms received a formal going-concern warning, while the remaining 830 did not.

ChatGPT identified 370 firms as “potential going concern warnings” by assigning them scores below 30. Of those 370, 297 were correct predictions (true positives), indicating that ChatGPT flagged 297 of the 320 actual going-concern firms. Meanwhile, 73 firms were flagged as at risk despite ultimately avoiding auditor warnings. These “false positives” (Type I errors) demonstrate some tendency of the model to err on the side of caution. By subtracting those 73 misclassifications from the 830 firms that remained stable, we arrive at 757 “true negatives” cases in which ChatGPT correctly recognised there was no going-concern issue. Finally, 23 firms slipped through ChatGPT’s classification as stable, or low-risk yet did in fact go on to receive going-concern warnings. These 23 “false negatives” (Type II errors) highlight the model’s failure to catch a small subset of truly distressed companies. A confusion matrix summarising these outcomes appears in [Table 4](#). While the model’s overall accuracy and F1 scores are strong, it is important to consider the practical ramifications of misclassifications in an SME context. A false positive occurs when a firm is flagged as distressed (going concern) despite ultimately remaining viable. This could lead management to divert resources toward crisis interventions unnecessarily, such as restructuring efforts or emergency capital-raising, potentially inhibiting growth or prompting undue alarm among investors and creditors.

Table 6

Regression analyses (marginal effects) of the probability of obtaining a going concern warning.

Dep. Var.	Going Concern Warning				
	Model 1 Marginal Effects	Model 2 Marginal Effects	Model 3 Marginal Effects	Model 4 Marginal Effects	Model 5 Marginal Effects
<i>Independent Variables</i>					
LLM Score	−0.0613*** (0.004)		−0.021*** (0.003)	−0.020*** (0.003)	−0.018*** (0.003)
LLM Red-Flag		0.512*** (0.008)			
Total Assets (Firm Size)			−0.001*** (0.000)	−0.001*** (0.000)	−0.001*** (0.000)
Revenue			−0.003*** (0.001)	−0.003*** (0.001)	−0.002*** (0.001)
Revenue (YoY Growth)			−0.008*** (0.001)	−0.007*** (0.002)	−0.007*** (0.002)
Profitability			−0.005*** (0.000)	−0.004*** (0.000)	−0.004*** (0.001)
Profitability (Margin)			−0.001* (0.000)	−0.001 (0.000)	−0.001 (0.000)
Retained Earnings			−0.012 (0.009)	−0.011 (0.009)	−0.011 (0.009)
<i>Control Variables</i>					
Firm Age				−0.002*** (0.000)	−0.002*** (0.000)
Gearing				0.003*** (0.001)	
Gearing_095					0.185*** (0.002)
Liquidity				−0.005*** (0.001)	
Liquidity_075					0.278*** (0.006)
Debt/Total Assets				0.008*** (0.002)	0.007*** (0.002)
Total Liabilities				0.001** (0.000)	0.001** (0.000)
Interest Coverage				−0.002* (0.000)	−0.002* (0.000)
Bank				−0.007 (0.005)	−0.006 (0.005)
Receivable Days				−0.001 (0.000)	−0.001 (0.000)
Payable Days				0.002 (0.000)	0.002 (0.000)
Sector					Yes
Location					Yes
# Obs.	4,600	4,600	4,600	4,600	4,600
LR-chi2	136.99	98.43	209.58	235.03	287.57
Pseudo-R2	0.4916	0.3298	0.3969	0.4510	0.4982

Notes: Table 6 presents the marginal effects result of probit regression model. The dependent variable is Going Concern Warning, a binary variable equals to 1 if a firm obtained a going concern warning, and 0 otherwise. Model 1 presents a simple baseline model with the LLM score (0–100). Model 2 replaces the LLM score with an LLM Red-Flag warning (binary) variable. Model 3 extends on Model 1 but expands on firm size and profitability metrics. Model 4 expands on Model 3 and includes more financial health metrics such as liquidity, gearing, interest coverage etc. Model 5 extends on Model 4 and includes gearing and liquidity thresholds as well as controlling for Sector and Location fixed effects. The data is obtained via FAME, and all variables are defined in Table 1. *** Statistical significance at the 1 % level. ** Statistical significance at the 5 % level. * Statistical significance at the 10 % level.

Equally, a false negative arises when a genuinely distressed SME goes undetected, representing a more severe risk: lacking advance warning, the firm may fail to enact timely remedial measures, which can hasten insolvency and potential job losses. Therefore, while the model's tendency to act with caution (73 false positives vs. 23 false negatives) is arguably preferable from a risk-avoidance standpoint, it highlights the importance of balancing the costs associated with each type of error. For SMEs with limited capital and managerial bandwidth, both Type I and Type II errors carry consequences that necessitate careful human oversight and strategic follow-up on AI-generated outputs.

Separately, precision, recall and F-1 scores were calculated using the confusion matrix to obtain an estimate on the accuracy of the potential going concern warning assessment and the actual outcomes in the years post-assessment. The overall accuracy of the model is calculated as the total correct classifications divided by the total companies. Table 5 summarises the results for reach category.

Precision, which gauges the reliability of flagged cases, reached 80.3 %, meaning that four out of every five firms classified as at risk ultimately did receive going-concern warnings. Recall, the model's ability to capture actual distressed firms, was 92.8 %, indicating it successfully identified most companies that were genuinely at risk. Balancing these two metrics, the F1-score came in at 86.2 %, reflecting strong performance in catching troubled SMEs while minimising false alarms. Finally, the model achieved an overall accuracy of 91.9 %, calculated as the total number of correct classifications divided by all firms in the sample. To contextualise ChatGPT's performance, we draw on established benchmarks in SME risk prediction. Traditional logit-based models typically report accuracies in the 80–90 % range (Hensher and Jones, 2007; Kristanti and Dhaniswara, 2023), though some mixed logit implementations have been optimised to reach comparable levels (Chang and

Lusk, 2011). More recently, Cheraghali and Molnár (2024) reviewed 145 SME default prediction models, noting that logistic regression methods commonly achieve around 82.9 % accuracy, whereas advanced machine learning algorithms (e.g., random forests, hybrid GBDT-CNN-LR) can exceed 99 % under certain conditions. Against this backdrop, ChatGPT's overall accuracy of 91.9 % is competitive with conventional econometric techniques and approaches the lower band of state-of-the-art machine learning results.

Notably, however, many of the top-performing machine learning models are highly specialised and may require substantial computational expertise or data preprocessing (Alaka et al., 2018; Pan et al., 2017). By contrast, ChatGPT's user-friendly interface and ease of integration could make AI-driven analysis more accessible to SMEs lacking advanced technical capabilities (Mollick and Euchner, 2023). While the model does not surpass every specialised solution reported in the literature, its balance of accuracy (91.9 %) and usability offers a compelling alternative for resource-constrained firms, potentially bridging the gap between cutting-edge analytics and everyday financial decision-making.

We also measure predictive accuracy using the Brier Score (Brier, 1950), which quantifies the mean squared difference between predicted probabilities and actual outcomes, with lower values indicating better performance. According to Roulston (2007) and Wilks (2011), a Brier Score of around 25 % is equivalent to random guessing ('no skill'); our model's score of 13.22 % therefore suggests that its predictions offer meaningful insight. However, there remains room for improvement as it is not completely 100 % accurate.

Fig. 2 below visually demonstrates how well the predicted probabilities align with the actual results via a calibration plot.

This calibration plot compares the average predicted probability per the LLM score to that of the averages of the going concern results at each

Table 7
Pearson correlation table.

	GCW	LLM Score	LLM Red-Flag	Total Assets	Revenue	Profitability	Firm Age	Gearing	Liquidity	Debt/Total Assets	Total Liabilities	Interest Cover	Bank	Receivable Days	Payable Days
Going Concern Warning (GCW)	1.0000														
LLM Score	-0.2593***	1.0000													
LLM Red-Flag	0.7284***	-0.7004***	1.0000												
Total Assets	-0.1467***	0.1523***	-0.1005***	1.0000											
Revenue	-0.1013***	0.1045***	0.7982***	1.0000											
Profitability	-0.2986***	0.0498	-0.1057***	0.4976***	1.0000										
Firm Age	-0.0524	0.1013***	-0.0518	0.2965***	0.2043***	1.0000									
Gearing	0.4539***	-0.0465	0.0587	0.1005***	0.0527	0.0527	1.0000								
Liquidity	-0.3956***	0.0521	-0.0624	0.1087***	0.0527	0.0527	0.1005***	1.0000							
Debt/Total Assets	0.2491	-0.1089***	0.1123***	0.2994***	0.0012	0.0492	0.0456	-0.3002***	1.0000						
Total Liabilities	0.3541	0.2457***	0.2457***	0.1034***	0.1034***	0.1554***	0.0999***	0.5031***	-0.4021***	1.0000					
Interest Cover	-0.0842***	-0.3053***	0.3821***	-0.1023***	-0.1023***	-0.2750***	-0.3053***	0.2457***	0.1034***	-0.1023***	1.0000				
Bank	-0.0742***	0.3821***	0.3225***	0.0108	0.0178	-0.1345***	0.3821***	-0.1345***	0.0123	-0.1345***	0.2457***	1.0000			
Receivable Days	-0.0820***	-0.1023***	-0.1023***	0.0450***	0.2205***	0.2205***	-0.1023***	0.0123	0.2424	0.2205***	0.1034***	-0.1023***	1.0000		
Payable Days	0.0742	-0.2750***	-0.1345***	0.2205***	0.1042	0.6031***	-0.2750***	-0.1345***	0.2205***	0.6031***	0.4521	-0.2750***	0.2205***	1.0000	

Notes: This table reports Pearson pairwise correlations. Correlations greater than approximately 0.06 in absolute value are significant at $p < 0.01$ (***). The data are from the FAME database, and all variables are defined in Table 1.

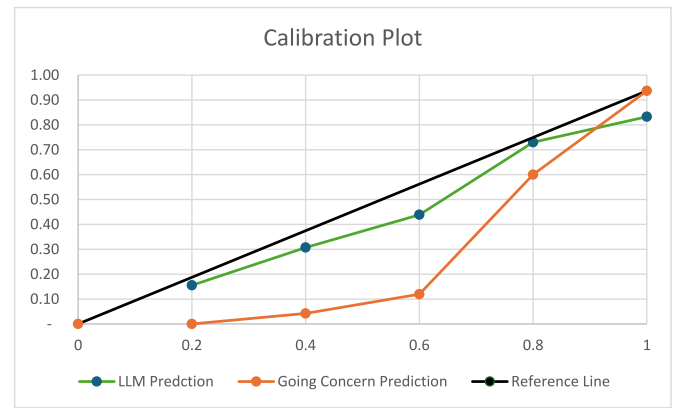


Fig. 2. Calibration plot of Brier score.

interval. We have chosen intervals of 20 % for the purpose of this study. The reference line per the graph shows perfect prediction, thus the closer to the reference line the more accurate the model. We can see that the LLM is highly accurate at correctly assessing companies that are clearly going concern risks and those that are safe. The highest distance per the graph is in relation to the 60 % chance of default interval. This indicating that the model is not perfect in its prediction. While the Briers score gives an overall assessment of the accuracy of the model, the calibration plot highlights the accuracy at different intervals.

4.2. Robustness check: probit analysis

In addition to evaluating ChatGPT's classification performance through accuracy metrics and confusion matrices, we conducted a Probit regression to statistically validate whether the LLM score remains a significant predictor of going-concern warnings after controlling for traditional financial indicators. Prior to undertaking our Probit regression, we use some diagnostic tools to assess for multicollinearity. Table 7 present a Pearson Correlation which is shows generally moderate pairwise correlations among the key variables, indicating minimal concern about severe multicollinearity. Although the correlation between Going Concern Warning (GCW) and LLM Red-Flag (0.7284***) is relatively high, this aligns with the fact that both are meant to capture the dimension of financial distress. In line with theoretical expectations, GCW is positively associated with Gearing (0.4539***) and negatively associated with Liquidity (-0.3956***), while the LLM Score is negatively correlated with GCW (-0.2593***), suggesting that higher AI-assessed financial stability corresponds to a lower incidence of going-concern problems. Overall, these results indicate that no single pairwise correlation is so large as to pose a major risk of collinearity in the subsequent regression analyses.

Concerning the Probit regression, our dependent variable is Going Concern Warning, a binary outcome coded as 1 if the firm received a warning in Years 5–8 (T+1 to T+4) and 0 otherwise. The primary independent variable is the LLM Score that ChatGPT assigned (scaled from 0 to 100). We first estimate a simple Probit model using only the LLM Score, then expand the specification to include conventional predictors such as financial health metrics and controlling for sector and location of firm. By comparing coefficients and significance levels across these models, we assess whether ChatGPT's risk assessment continues to exhibit incremental predictive value beyond standard financial metrics. This approach serves as a robustness check, further illustrating how an AI-generated score can complement or enhance SME financial diagnostics.

The results from the Probit regression analyses reveal that the LLM Score remains a significant predictor of a firm's likelihood of receiving a going-concern warning, even after controlling for a variety of traditional financial variables. Specifically, the LLM Score displayed a strongly

negative marginal effect, with a one-point increase in the score leading to a decline in the probability of receiving a going-concern warning by approximately 6.13 percentage points (Model 1). This suggests that firms with higher LLM Scores (indicating lower risk as assessed by the AI model on a score of 0–100) are significantly less likely to be flagged for potential going-concern issues. In Model 2, when the LLM Score is transformed into a binary red-flag variable (for firms flagged as high-risk by ChatGPT), the analysis reveals a positive marginal effect, with firms labelled as “red flags” exhibiting a 51.2 percentage-point higher likelihood of receiving a going-concern warning.

In Model 3, the LLM Score retains its negative and statistically significant relationship with going-concern warnings after additional financial controls are introduced, indicating that firms with higher AI-generated scores continue to show a lower probability of receiving a warning. Model 4 adds further balance-sheet variables, yet the LLM Score remains robustly significant and negatively signed, highlighting its persistent predictive value even as more comprehensive financial indicators are incorporated. Finally, Model 5 incorporates specific liquidity thresholds and adds sector and location fixed effects; although these additional controls do not yield statistically significant effects themselves, the coefficient for the LLM Score again remains negative and significant, highlighting its consistent explanatory power across all models.

5. Discussion and conclusion

This study set out to examine whether an LLM, specifically ChatGPT, could serve as an effective means of flagging potential going-concern issues in SMEs. By framing our investigation within a dual-theoretical perspective that integrates the RBV (Barney, 1991; Peteraf, 1993) with technology affordance and constraint theory (Majchrzak and Markus, 2013), we have been able to assess how AI tools can extend the capabilities of resource-constrained firms while simultaneously introducing new limitations, particularly regarding interpretability and data quality. The results from both classification performance and probit analyses show that ChatGPT’s LLM score meaningfully contributes to predicting going-concern warnings beyond conventional financial metrics, reinforcing the notion that AI-based screening can help SMEs identify red flags earlier than standard audits or internal reviews alone (Mai et al., 2019; du Jardin, 2021).

Viewed through the lens of technology affordance and constraint theory, ChatGPT offers clear affordances that can benefit resource-constrained organisations. First, the model’s rapid and relatively low-cost analysis of large numerical datasets enables managers to quickly identify financial red flags that might otherwise go undetected until formal audits take place. This accessibility advantage can be especially impactful for SMEs lacking the capital or in-house expertise to deploy sophisticated credit-scoring systems or predictive analytics platforms (Alaka et al., 2018). Additionally, the user-friendly interaction of a chat interface allows managers with varying degrees of financial literacy to gather insights about their firms’ vulnerabilities without requiring specialised statistical or coding skills.

While technology affordances and constraints emphasise that any new technological tool introduces its own set of limitations, AI-driven solutions such as ChatGPT in financial diagnostics are no exception. The inherent black-box nature of LLMs, while enabling fast and automated analysis, could pose challenges in terms of interpretability. Even though our findings confirm that ChatGPT is highly accurate, its recommendations demand careful interpretation and prompt action. Entrepreneurs must allocate time to understand these outputs and decide when to intervene. Therefore, the effectiveness of ChatGPT’s insights depends not only on advanced system inputs but also on the ability of managerial teams to integrate these signals judiciously into their broader decision-making frameworks. These observations also echo prior scholarship indicating that technology alone cannot ensure improved managerial decision-making without complementary

organisational adaptations and capabilities (Freel, 1999; Carter and Van Auken, 2006). Indeed, our results confirm that while tools like ChatGPT can offset SMEs’ resource constraints, success depends on entrepreneurs’ readiness to interpret and act on AI-derived insights (Maragno et al., 2023).

Integrating the RBV with technology affordance and constraint theory further enriches our understanding of these dynamics. The RBV reminds us that SMEs are often at a disadvantage because they lack the internal resources needed for sustained competitive advantage (Hitt et al., 2001). In this context, LLMs provide an external digital asset that can help bridge this resource gap. However, as technology affordance theory warns, the successful adoption of such tools depends not only on their intrinsic capabilities but also on the readiness of the firm to integrate and effectively interpret the outputs. Thus, our findings imply that while LLMs have the potential to revolutionise financial risk assessment, their true value is realised only when SMEs invest in the complementary capabilities, such as digital literacy and managerial insight, that are essential for leveraging these advanced technologies.

From a theoretical standpoint, these observations broaden the scope of technology affordance and constraint theory within the context of AI-driven predictive tools. ChatGPT’s capacity to consolidate vast amounts of numerical data into an easily understandable probability of distress introduces a novel affordance, a user-friendly, efficient, and proactive risk assessment tool. On the other hand, this same capability highlights constraints, such as the dependency on sophisticated programming or inputs, the need for immediate and accurate interpretation of signals, and the risk that managers might either disregard important warnings due to a lack of insight into the model’s reasoning or overly depend on automated outputs without contextual judgment.

From a practical perspective, our findings suggest that SMEs could adopt ChatGPT-like solutions as a cost-effective “early warning system.” For policymakers and accountants, the study highlights an opportunity to integrate AI into auditing workflows or external monitoring methods, potentially improving the timeliness and accuracy of risk detection across the broader SME sector. Meanwhile, AI developers could focus on creating interpretability features tailored to small business users, such as explanations or diagnostic summaries that clarify why a firm is flagged, thereby helping overcome distrust in unknown algorithms.

Although our findings demonstrate the tangible benefits of ChatGPT as an early-warning system, they also highlight the need for robust ethical and governance frameworks, particularly in smaller firms that lack formal governance structures such as ethics committees or compliance teams. Existing AI ethics guidelines, such as those surveyed by Hagendorff (2020) and Heiling (2022), frequently share core principles like transparency, accountability, and fairness but often fail to translate these into actionable standards. This is especially problematic in an SME context, where resource limitations can push aside careful oversight in favour of immediate efficiency gains. Recent work on AI governance in the private sector (Schneider et al., 2023; Birkstedt et al., 2023) proposes firm-centric frameworks that highlight data quality, model oversight, and stakeholder engagement. However, SMEs face practical hurdles in implementing these frameworks, such as the time and financial costs of compliance and a lack of internal AI expertise.

An additional challenge stems from interpretability. While ChatGPT’s text-based interface lowers technical barriers, entrepreneurs may not fully grasp how the model arrives at its predictions or what biases might lurk within the training data (Ryan, 2020). Consequently, SMEs adopting ChatGPT-based solutions must couple them with human oversight, ensuring decision-makers critically evaluate and contextualise each output rather than deferring entirely to algorithmic recommendations (Jobin et al., 2019; Corrêa et al., 2023). A more formal AI governance structure, adapted from large-firm best practices, yet scaled to smaller operations, could help entrepreneurs address these concerns. Although ChatGPT’s conversational interface lowers the technical threshold, SMEs may still face financial and operational barriers in deploying AI solutions. Subscription fees, hardware

requirements, and the need for stable data pipelines can pose formidable challenges for smaller enterprises (de Almeida et al., 2021). Additionally, owners or managers must learn to interpret model outputs accurately and allocate sufficient time to integrate AI-driven insights into strategic decisions. Adopting a phased rollout, starting with pilot projects or partnering with external service providers or their small-to-medium sized accounting practitioners, could help SMEs minimise risk and gain familiarity with generative AI workflows (Schneider et al., 2023). While generative AI tools hold great promise in augmenting financial risk management for resource-constrained firms, their success hinges on accessible implementation pathways, supportive organisational culture, and an ongoing commitment to ethical governance.

Despite these promising results, there are important limitations that pave the way for future research. Although the LLM draws meaning from structured financial data, unstructured information, such as narrative disclosures or managerial communications, remains largely unexplored and might further refine the model’s predictive strength. Future investigations could incorporate textual data from management reports, test real-time continuous monitoring approaches, or conduct user acceptance studies that evaluate how managers integrate AI recommendations into routine financial decision-making (Manigart and Khosravi, 2024). Further research might also explore the applicability of ChatGPT-like tools to broader managerial accounting processes and budgetary planning, where early identification of cost overruns or revenue shortfalls could prevent crises from escalating. It should also be noted that our analysis was confined to ChatGPT o1 reasoning at a specific point in time, and given the rapid evolution of LLM’s, future versions or alternative AI tools may yield different outcomes or deeper insights. A key limitation is ChatGPT’s proprietary and evolving nature, which may affect reproducibility of our exact results over time, highlighting the need for ongoing validation with open-source or updated

models.

In conclusion, by bridging the resource gap for small businesses through the integration of RBV and technology affordance and constraint theory, our study provides empirical evidence that LLMs such as ChatGPT can offer valuable assistance to managers in overseeing their financial health. This finding not only highlights the potential of AI solutions in mitigating business and financial risks among SMEs but also contributes to the growing body of literature on how digital tools can be systematically harnessed to enhance SMEs’ resilience and strategic agility. The connection between resource constraints and digital affordances is key to understanding how SMEs can successfully adopt AI-driven methods, ultimately driving more informed, data-driven decision-making that improves long-term viability.

CRediT authorship contribution statement

Michael Metzger: Conceptualization, Software, Data curation. **Seán O’Reilly:** Writing – original draft, Investigation, Supervision, Formal analysis, Writing – review & editing, Methodology, Conceptualization. **Ciarán Mac an Bhaird:** Writing – review & editing, Conceptualization, Supervision.

Declaration of competing of interest

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Michael Metzger | Seán O’Reilly | Ciarán Mac an Bhaird.

Appendices.

Appendix 1Scoring System

Variable	Criteria	Points
1. Year-on-Year (YoY) Revenue Growth	<0 % growth	0–4
	0–10 % growth	5–7
	≥10 % growth	8–10
2. Profit Margin	Negative margin in any of the last 2 years	0–6
	Positive but <5 % margin overall	7–11
	≥5 % margin on average	12–15
3. Liquidity Ratio (Current Ratio)	<1.0	0–6
	1.0–1.5	7–11
	≥1.5	12–15
4. Gearing Ratio (Debt/Equity)	≥2.0	0–4
	1.0–2.0	5–7
	<1.0	8–10
5. Debt-to-Total-Assets	≥0.8	0–4
	0.5–0.8	5–7
	<0.5	8–10
6. Interest Coverage (EBIT/Interest Expense)	<1.0	0–4
	1.0–2.0	5–7
	≥2.0	8–10
7. Receivable Days	≥60 days	0–4
	30–60 days	5–7
	<30 days	8–10
8. Payable Days	<30 days	0–4
	30–60 days	5–7
	≥60 days	8–10
9. Bank Balance	<1 month of operating expenses	0–4
	1–3 months	5–7
	≥3 months	8–10

Appendix 2. Scoring Classification

Total Score Range	Risk Classification	Description
0–30	High Risk (Red)	Indicates a high likelihood of going-concern issues or severe financial distress.
31–50	Moderate Risk (Amber)	Indicates moderate caution; the firm may be vulnerable to financial shocks.
51–75	Stable (Green)	Indicates stable financial performance and resilience.
76–100	Strong (White)	Indicates strong financial health and robust capacity to handle any short-term challenges.

Appendix 3. Sample Prompt

“Based on the following company profile and four-year financial history, please evaluate the company’s risk of going-concern issues and assign an overall LLM Score between 0 and 100. Use the scoring system provided (Appendix 1) to guide your analysis, and conclude with a risk classification as follows:”

Overall Scoring

- High Risk (Red): Total Score 0–30
- Moderate Risk (Amber): Total Score 31–50
- Stable (Green): Total Score 51–75
- Strong (White): Total Score 76–100

“Also, provide a red flag indicator (1 if the LLM Score falls between 0 and 30, 0 otherwise)”

Company Profile:

Sector [Sector Classification: e.g., Business and Professional Services, Construction and Property, Consumer Services, Manufacturing, Retail and Wholesale]

Location: [London, South, East, North, West, or Other UK]

Firm Age: [Number of years since incorporation]

Four-year Financial History of Company 1 (Illustration purposes).

No.	Financial Metric	Prompt
1	Revenue and YoY Growth:	Year 1: Revenue = [Value] (log of sales in £), YoY Growth = [Value]% Year 2: Revenue = [Value] (log of sales in £), YoY Growth = [Value]% Year 3: Revenue = [Value] (log of sales in £), YoY Growth = [Value]% Year 4: Revenue = [Value] (log of sales in £), YoY Growth = [Value]%
2	Profitability and Profit Margin:	Year 1: Profitability = [Value] (log of EBITDA in £), Profit Margin = [Value]% Year 2: Profitability = [Value] (log of EBITDA in £), Profit Margin = [Value]% Year 3: Profitability = [Value] (log of EBITDA in £), Profit Margin = [Value]% Year 4: Profitability = [Value] (log of EBITDA in £), Profit Margin = [Value]%
3	Total Assets:	[Log amount of total assets]
4	Retained Earnings:	[Log amount of retained earnings at financial year-end]
5	Gearing:	Gearing Ratio (Debt/Equity): [Value] Gearing.095 (Dummy: 1 if gearing ratio >0.95, otherwise 0): [Value]
6	Liquidity:	Liquidity Ratio (Current Assets/Current Liabilities): [Value] Liquidity.075 (Dummy: 1 if liquidity ratio <0.75, otherwise 0): [Value]
7	Debt/Total Assets:	[Value]
8	Total Liabilities:	[Log amount of total liabilities]
9	Interest Coverage:	[Value] (EBIT/Interest Expense)
10	Bank Balance:	[Log amount of bank balance (£)]
11	Receivable Days:	[Value] days (Calculated as: Trade Receivables/Revenue × 365)
12	Payable Days:	[Value] days (Calculated as: Trade Payables/Cost of Sales × 365)

Instructions: Please analyse each key financial metric according to the scoring criteria above. Summarise your evaluation by calculating an overall score (LLM Score) between 0 and 100. Conclude with the company’s risk classification based on the total score and include a red flag indicator (assign a dummy value of 1 if the overall score is between 0 and 30, indicating red flag, and 0 otherwise).

Data availability

Data will be made available on request.

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